SprayBench: Benchmarking LLMs on Open-Ended Materials Reasoning in Additive Manufacturing

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Abstract

While LLMs are increasingly used in scientific workflows, their ability to support open-ended manufacturing decisions involving material systems, processing conditions, process-property tradeoffs, and sparse experimental evidence remains unclear. We introduce **SprayBench**, a 525-question benchmark for open-ended reasoning in the emerging field of cold spray materials. SprayBench combines source-anchored literature extraction with cold-spray expert refinement to create practitioner-style questions targeting explanation, tradeoff analysis, and process guidance. Across 34 frontier and open-weight configurations, with open-weight models run locally on a DGX Spark, we evaluate closed-book performance, source-controlled RAG on 13 configurations, and LLM-as-a-judge reliability. We find that, under closed-book inference, open-weight pass rates range from 6.3% to 65.8%, while the strongest frontier model reaches 85.1%. Source-excluded RAG improves all tested configurations by +3.1 to +29.4 percentage points, even when the originating document is withheld. Repeated LLM judging and a 105-response stratified expert audit quantify reviewer noise, model drift, and pass-fail threshold effects in open-ended grading.

1 Introduction

Large language models (LLMs) are increasingly used in scientific and engineering workflows, including materials literature mining, predictive modeling, tool calling, and agentic systems [1, 10, 12, 19, 22]. However, evaluating whether these systems can support process-specific materials decisions remains difficult. General-purpose benchmarks provide limited evidence for specialized materials workflows. Recent materials and chemistry benchmarks are more domain-relevant, but often target broad domain knowledge, literature-grounded QA, multimodal problem solving, or computational tool use [2, 5, 10, 18, 20]. However, these benchmarks do not directly test manufacturing guidance requiring models to connect process variables, mechanisms, material constraints, and property tradeoffs. In real-world applications, even a basic choice, such as selecting helium or nitrogen as the carrier gas, can require balancing particle acceleration, heat input, oxidation risk, material compatibility, and cost against sparse experimental evidence and equipment-specific process knowledge. As a result, strong performance in current benchmarks can hide failures that would arise in open-ended technical responses essential to support real-world engineering decisions [6, 13].

We study this gap through cold spray, a solid-state coating and additive-manufacturing process used to restore parts, extend service life, and reduce repair downtime and costs [3]. Cold-spray outcomes depend on complex process-structure-property relationships, such that carrier gas, powder material, particle temperature and velocity, substrate condition, and post-treatment jointly shape deposition efficiency, microstructure, and mechanical performance. As a result, useful cold-spray guidance requires reasoning across coupled constraints rather than treating individual trends in isolation.

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Additionally, because information is fragmented across experimental studies and machine-readable datasets remain limited [12], cold spray provides a test case for evaluating whether models can ground process-specific guidance in sparse, distributed literature rather than curated datasets.

This setting raises *three evaluation challenges*. First, a useful benchmark must provide coverage over interconnected and practitioner-aligned mechanisms, process variables, material systems, and property outcomes rather than isolated facts. Second, closed-book failures may reflect missing domain knowledge rather than poor reasoning. Retrieval can mitigate this but must be evaluated under document-level source control to separate broad literature grounding from direct access to the QA-construction source. Third, open-ended answers cannot be scored by exact match, while exhaustive subject-matter-expert (SME) review is impractical. LLM-as-a-judge evaluation can scale grading but requires calibration for reviewer drift, decision-boundary effects, and expert agreement.

To address these challenges, we present three contributions:

- We introduce **SprayBench**, a 525-question benchmark for open-ended process-structure-property reasoning in cold spray. SprayBench combines source-anchored literature extraction with cold-spray expert refinement to produce practitioner-style questions targeting explanation, tradeoff analysis, and process guidance. *The full SprayBench question and target-answer set, as well as candidate-model outputs, can be found at https://github.com/sprice134/SprayBench.*
- We evaluate 34 frontier and open-weight configurations under closed-book inference and 13 under source-controlled RAG, comparing source-excluded and source-eligible retrieval to distinguish broad literature grounding from direct access to the originating source.
- We further calibrate scalable open-ended grading using repeated judge reviews, comparison with the larger `GPT-5.4`, several open-weight reviewer ablations, and a 105-response stratified expert audit blind to model identity and LLM scores.

2 SprayBench Construction and Evaluation Protocol

Benchmark construction. SprayBench was constructed using a two-stage human-in-the-loop pipeline. First, `GPT o4-mini` [11] parsed cold-spray and additive-manufacturing literature to extract source-anchored question-answer pairs. Second, a Ph.D.-level cold-spray expert reviewed each candidate pair, correcting errors, adding missing context, and aligning questions toward practitioner-facing explanations, tradeoff analysis, and process guidance. Answer-provenance analysis shows that SME refinement expanded the LLM-extracted answers while retaining 46.6% of their original tokens (Appendix A). The final benchmark contains 525 SME-refined QA pairs from 444 source documents across three categories: process configuration, materials mechanisms, and mechanical performance. Questions span three intended complexity levels, from direct process trends to multistep process-structure-property reasoning (Appendix A.1.1).

For example, a Level 2 SprayBench question asks: *"How do differences in drag acceleration and inertia between small and large particles create a characteristic maximum particle velocity within a specific size range in low-pressure cold gas spraying?"* The SME-refined target-answer explains how small particles accelerate rapidly in the nozzle but decelerate near the substrate and large particles accelerate less efficiently due to their greater inertia, while intermediate-size particles reach the highest impact velocities. The complete question-answer pair is provided in Appendix A.1.2.

Model inference. We evaluate 34 model configurations, including four API-accessible frontier models and 30 open-weight configurations spanning Qwen, Gemma, Nemotron, GPT-OSS, Magistral, and Granite. API-accessible models, all queried through official provider endpoints, serve as external performance references. Open-weight models are hosted on a single DGX Spark to reflect compute-constrained and privacy-compatible materials research settings. All models use identical prompts. Local inference uses LM Studio/`llama.cpp`, GGUF weights, a temperature of 0.0, each model's maximum supported context length, and `Q4_K_M` or the nearest available 4-bit quantization. Complete model identifiers, inference settings, and parameter metadata are reported in Appendix B.

Retrieval protocol. To evaluate retrieval under document-level source control, we index 5,205 cold-spray and additive-manufacturing articles into 80,161 chunks. For each question, `BAAI/bge-m3` retrieves 40 candidate chunks, `BAAI/bge-reranker-v2-m3` reranks them, and the top 8 eligible

chunks are included as context. We compare closed-book inference against two RAG conditions: source-excluded and source-eligible. In the source-excluded condition, all chunks from the question's source document are filtered out prior to context selection. In the source-eligible condition, that document is available for retrieval but not guaranteed to appear in the final context. Prior to question generation, articles were de-duplicated using title and DOI to reduce potential source leakage. However, alternate versions, such as preprints with different titles or DOIs, may remain eligible for retrieval. Retrieval diagnostics and source-identification analyses are provided in Appendix C.

Evaluation protocol. Because SprayBench answers are open-ended and require domain-specific knowledge, exact-match and similarity-based scoring are insufficient. We therefore use a reference-guided, rubric-based LLM-as-a-judge protocol. Each candidate response is scored against the SME-refined target using three independent `GPT-5.4-mini` reviews. Each review produces a binary pass/fail decision and 0-5 scores for *factual accuracy*, *completeness*, *specificity*, *relevance*, *absence of misconceptions*, and *overall quality* (Appendix D.1). Binary decisions are aggregated by majority vote and numeric scores are averaged across reviews. We calibrate this protocol using repeated judge reviews, comparison between `GPT-5.4-mini` and `GPT-5.4`, open-weight reviewer ablations, and a blinded 105-response stratified expert audit. Complete inference, RAG, and judge prompts are provided in Appendix F. We further evaluate cost-aware judge routing in Appendix E.4.

3 Results

Closed-book Performance. Evaluating closed-book inference establishes a baseline for out-of-the-box model performance on process-specific materials questions. Across 34 evaluated model configurations, open-weight models ranged in pass rate from 6.3% to 65.8%, while the strongest API-accessible frontier model achieved 85.1%, as shown in Figure 1. The strongest locally hosted models, `Qwen3.6-27B` and `Gemma4-31B`, approached `o4-mini` and `Gemini-3-Flash-Preview`, although the strongest frontier model still retained a significant performance lead. Because output verbosity can bias LLM-as-a-judge evaluations [21], we also examined model-level factors associated with SprayBench performance. Total parameters, active parameters, and reasoning-token length correlated with pass rate, whereas final-answer length did not (Appendix E.3.2). Within GPT-OSS, moving from low to high reasoning reduced closed-book pass rates for the 20B and 120B models by 2.9 and 2.5 pp, respectively, but increased source-excluded RAG performance by 3.5 and 4.0 pp and source-eligible RAG performance by 1.3 and 1.9 pp (Appendix E.5.2).

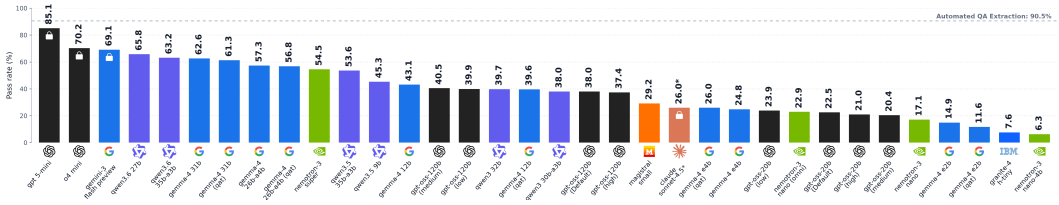


Figure 1: **SprayBench closed-book capability landscape.** Closed-book pass rates across 34 model configurations on 525 SME-refined open-ended questions, comparing open-weight models with API-accessible frontier baselines (marked by lock icons). The dashed line marks the automated QA extraction baseline (90.5%), and the asterisk indicates Claude's refusal-inclusive score.

Source-controlled retrieval improves performance. As shown in Table 1, source-excluded RAG improved performance across all seven models, with an average gain of +15.5 percentage points and a maximum gain of +25.4 pp. Gains were largest for weaker models, but even `GPT-5-mini`, the strongest baseline, improved by 3.1 pp. Source-eligible retrieval increased performance further, but additional gains were smaller, averaging only +3.7 pp.

Across 6,808 matched model-question pairs with closed-book and source-excluded responses, spanning seven unique models and six additional GPT-OSS reasoning configurations, source-excluded RAG changed acceptability on 2,262 responses, with 1,802 closed-book failures becoming acceptable and 460 closed-book passes becoming rejected. Paired McNemar tests showed a positive rescue-harm imbalance that was statistically significant for most configurations, indicating dataset-level gains despite occasional question-level degradation. Under source-eligible retrieval, the question's originating source document appeared in the final top-8 context for 27.1% of matched questions. On source-hit

Table 1: **Matched RAG pass rates on representative configurations.** Values are pass rates (%); parentheses show gain over the matched closed-book result.

Condition	GPT-5 mini	Qwen3.6 35B A3B	Gemma4 26B-A4B	Nemotron3 Super	GPT-OSS 120B	Gemma4 E4B	GPT-OSS 20B
Base performance	85.1	63.3	57.3	54.5	38.0	24.8	22.5
Source-excluded RAG	88.2 (+3.1)	73.2 (+9.9)	61.1 (+3.8)	71.9 (+17.4)	62.2 (+24.2)	49.8 (+25.0)	47.9 (+25.4)
Source-eligible RAG	91.4 (+6.3)	77.2 (+14.0)	65.7 (+8.4)	74.8 (+20.3)	68.3 (+30.3)	51.9 (+27.1)	51.0 (+28.4)

questions, pass rate increased by 14.5 pp over source-excluded RAG, compared with -0.2 pp for non-hit questions.

Evaluator reliability and calibration. Open-ended answers better align with practitioner requirements but cannot be scored by exact match or symbolic verification, motivating LLM-as-a-judge evaluation. Thus, in addition to candidate-model performance, we evaluated reviewer-model reliability and calibration. Repeated reviews reduced stochastic scoring noise, with averaging three `GPT-5.4-mini` reviews increasing overall-score ICC from 0.912 to 0.968, while three `GPT-5.4` reviews increased ICC from 0.935 to 0.977. However, high intra-judge reliability did not guarantee inter-judge agreement. Using three-review average scores, `GPT-5.4-mini` and `GPT-5.4` showed strong score agreement ($\mathrm{ICC} = 0.874$), but pass-fail labels disagreed on 12.3% of responses. These disagreements were concentrated near the decision boundary and strongly directional, with 91.2% occurring when `GPT-5.4-mini` accepted a response that `GPT-5.4` rejected.

Despite this response-level leniency, average overall-score rankings remained strongly aligned ($\rho = 0.994$, $\tau_b = 0.950$). Using these findings, *SprayRouter* escalates uncertain responses to stronger judges, preserving 98.1% agreement with the three-review `GPT-5.4` reference while reducing full-judge costs by 69.0%. Additional reviewer ablations, including open-weight judges, showed larger directional leniency, reinforcing that reviewer models must be calibrated rather than treated as interchangeable. A stratified expert audit showed score-level alignment with human scoring but stricter LLM acceptability thresholds, with nearly all disagreements being human-accepted responses rejected by LLM reviewers. Full reviewer-alignment, open-weight-judge, and expert-audit analyses are provided in Appendix E.2, with judge-routing analyses in Appendix E.4.

4 Discussion and Conclusion

In this work, we present **SprayBench**, a 525-question benchmark for open-ended process-structure-property reasoning in cold spray, where useful answers require explanation, tradeoff analysis, and process guidance grounded in material systems, operating conditions, and experimental evidence. Across 34 frontier and open-weight configurations, SprayBench shows that open-ended cold-spray reasoning remains challenging. Locally hosted open-weight models, necessary for privacy- and information-regulated materials workflows, are competitive with several API-accessible frontier baselines. However, the strongest frontier model still outperformed the best open-weight model, making effective private, locally deployable materials reasoning an important research topic.

These results show that source-controlled RAG improves open-ended cold-spray answering. In source-excluded RAG, where the question's source document is removed before context selection, RAG improved all seven model configurations, with an average pass-rate gain of +15.5 percentage points. Allowing the source document to be eligible for retrieval produced an additional, but smaller, gain. This suggests that direct source access can improve results when retrieved, but most of the retrieval benefit comes from broader literature grounding across distributed information sources.

SprayBench shows that open-ended materials evaluation is not only a model-inference task. LLM-as-a-judge protocols make large-scale grading feasible, but stochastic reviewer noise, model-to-model drift, and acceptability-threshold differences require evaluating candidate models and judge models together. We use cold spray as a focused case study because effective answers must connect process-specific cues, material constraints, property outcomes, and fragmented experimental evidence. This makes SprayBench a practical benchmark for both LLM-assisted manufacturing guidance and scalable evaluation of open-ended technical reasoning. Future work should extend SprayBench to other materials domains and manufacturing processes, expand independent expert review, and further calibrate local and routed judges.

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A Benchmark Construction and Answer Provenance

A.1 Construction Pipeline

SprayBench was constructed using a two-stage human-in-the-loop pipeline. First, OpenAI's `o4-mini` [11] parsed cold-spray and additive-manufacturing literature to extract source-anchored question-answer pairs. This extraction step was designed to increase coverage across mechanisms, process variables, material systems, microstructural outcomes, and mechanical-performance relationships. Next, a Ph.D.-level cold-spray expert reviewed each candidate pair, correcting erroneous or study-specific claims, adding missing technical context, and aligning the final questions and target answers toward practitioner-facing use cases, including mechanistic explanation, tradeoff analysis, and process guidance.

The final 525-question benchmark spans 444 unique source documents. Most sources contributed only one question, with an average of 1.18 questions per source, a median of 1.0, and a maximum of 3.0, providing a diverse set of questions across varying cold spray sub-topics.

A.1.1 Question Taxonomy and Examples

SprayBench questions are organized into three cold-spray-grounded categories, *Process Configuration*, *Materials Mechanisms*, and *Mechanical Performance*, and three intended complexity levels. These categories separate setup choices, underlying mechanisms, and property-level outcomes. *Process Configuration* questions focus on parameter selection and setup choices, including carrier gas, gas temperature and pressure, nozzle design, stand-off distance, particle velocity, and deposition efficiency. *Materials Mechanisms* questions focus on deformation, bonding, microstructural evolution, oxidation, grain refinement, and instability mechanisms. *Mechanical Performance* questions focus on coating properties and structural behavior, including hardness, strength, ductility, and fatigue crack growth. Together, these categories represent many of the process-structure-property decisions needed to support automated materials discovery and manufacturing workflows.

Table 2: **SprayBench candidate question taxonomy**. Breakdown of targeted candidate questions by category and intended complexity.

Category	Level 1	Level 2	Level 3	Target Total
Process Configuration	100	50	25	175
Materials Mechanisms	100	50	25	175
Mechanical Performance	100	50	25	175
Total	300	150	75	525

As shown in Table 2, each question is assigned an intended complexity level based on the number of reasoning steps expected for an effective answer. Level 1 questions target basic definitions, terminology, and direct process trends. Level 2 questions require linking process parameters or post-treatments to mechanisms and outcomes. Level 3 questions require multistep process-structure-property reasoning across interacting conditions, material systems, and performance metrics, such as using stress intensity factor formulation and Paris law to explain how increased specimen thickness from a cold spray coating can reduce fatigue crack growth rate.

This taxonomy also supports targeted failure analysis. For example, Level 3 questions were designed to require more reasoning steps, yet 24 of the 34 evaluated LLM configurations performed better on Level 3 than on Level 2 questions. This suggests that intended human complexity does not always map directly onto empirical model difficulty. In some cases, more complex questions may provide explicit analytical structure, terminology, or governing equations that help constrain an LLM response.

A.1.2 Representative Question-Answer Pair

To provide an example of the questions in SprayBench, we present a sample Level 2 materials-science question and its complete SME-refined target answer below.

- **Question (Q320, Level 2):** How do differences in drag acceleration and inertia between small and large particles create a characteristic maximum particle velocity within a specific size range in low-pressure cold gas spraying?
- **SME-refined target answer:** In low-pressure cold spray a peak in impact velocity appears at an intermediate particle size because of two opposing effects. Small particles, with little mass and a high drag-to-mass ratio, accelerate quickly in the nozzle and can nearly reach the gas velocity, but on exiting they have little inertia, so they are strongly decelerated in the compressed, stagnant gas layer just above the substrate. Large particles have more inertia, which resists the gas drag and keeps them from accelerating to high velocity in the nozzle in the first place. So the smallest are over-decelerated near the surface and the largest are under-accelerated in the nozzle, and the highest impact velocity occurs for particles of intermediate size. This velocity maximum produces a separation effect in the spray and narrows the deposition window, making the impact velocity, and so deposition, sensitive to the spread of particle sizes.

A.2 Evaluation of Stage 2 Human-in-the-Loop Refinement

To quantify the impact of expert refinement, we compare three answer sources: the initial source-anchored extraction, a question-only `o4-mini` response generated without literature grounding, and the final SME-refined target. For overlap and edit distance analyses, the Stage 1 extracted answer serves as the baseline. As shown in Table 3, the refinement stage was more than simple validation. The expert refinement expanded and corrected the extracted answers, ensuring that candidate models were evaluated against SME-refined technical targets rather than unreviewed LLM outputs. For example, automatically extracted answers contained 91.4 words on average, while SME-refined targets contained 141.1 words, reflecting the added context, tradeoffs, and practitioner-relevant detail. Although this process required significant human-effort, it did not require writing the benchmark from scratch. On average, 46.6% of tokens from the extracted answers were retained in the final targets, indicating that source-anchored extraction provided a strong draft for scaling benchmark construction.

Table 3: **Origin-anchored answer provenance.** Comparison between original extracted answers, free-form `o4-mini` answers, and final SME-refined targets.

Answer Source	Word Count	Mean Characters	TF-IDF Cosine	LSA Cosine	ROUGE-L	Bigram Jaccard	Insertion (%)	Deletion (%)
Origin-o4mini	91.4	6.17	–	–	–	–	–	–
o4-mini	294.8	5.41	0.183	0.653	0.138	0.029	93.0	79.0
SME target	141.1	5.56	0.396	0.806	0.378	0.130	69.8	53.4

The same pattern appears in standard natural language processing (NLP) similarity metrics. Compared with question-only `o4-mini` answers, SME-refined targets were closer to the source-anchored extractions across TF-IDF cosine, LSA cosine, ROUGE-L, and bigram Jaccard similarity. Comparing individual metrics, the SME target answer had a moderate lexical overlap with the extracted answer, with a significantly stronger semantic overlap, suggesting that refinement often changed wording, added context, and reorganized content while preserving the core scientific meaning. In comparison, question-only `o4-mini` responses were much longer and showed lower similarity and overlap with the source-anchored extractions.

We also evaluated the initial literature-grounded extractions and question-only `o4-mini` answers against the final SME targets using the same `GPT-5.4-mini` rubric used for candidate-model scoring. Under this LLM-as-a-judge evaluation, automated extractions achieved a 90.5% pass rate and an average overall score of 4.42, while question-only `o4-mini` outputs achieved a 70.2% pass rate and an average overall score of 3.64. This highlights how, in addition to enforcing QA diversity, literature grounding improved answer fidelity compared to the model closed-book output, with the largest gains appearing in factual accuracy (4.62 vs. 3.68) and absence of misconceptions (4.56 vs. 3.33).

B Model Inference Configurations

We evaluate 34 model configurations, including 30 open-weight configurations and four API-accessible frontier models. Open-weight models were hosted locally on a single DGX Spark to evaluate performance under compute-constrained and privacy-compatible deployment conditions. API-accessible models, including `GPT-5-mini`, `o4-mini`, `Gemini-3-Flash-Preview`, and `Claude-Sonnet-4.5`, were accessed through official provider endpoints.

B.1 Model Execution

To reduce inference variability, all open-weight models were run on the same hardware and local inference stack. Each model was evaluated in LM Studio using the `llama.cpp` backend, `GGUF` model containers, temperature `0.0`, and the maximum available context length for the loaded model. Despite fixed weights and temperature `0.0`, small nondeterminism was observed across LM Studio versions. Therefore, all reported local experiments used `LM Studio v0.4.16`. Each model used `Q4_K_M` quantization when available, and the nearest supported 4-bit format otherwise. For `GPT-OSS` models, we report results under the default reasoning configuration, as well as `low`, `medium`, and `high` reasoning settings.

B.2 Model Selection

Table 4 reports the models used throughout this analysis, including release year, provider, access type, active and total parameter counts, and input modality. Motivated by lab deployment settings where enterprise-scale compute may not be available, the primary open-weight configurations were restricted to models that could be loaded and run on a single DGX Spark. All selected models were released in 2025 or 2026. To enable model-factor analysis, the evaluated models span multiple families, including Gemma, Qwen, GPT-OSS, Nemotron, Magistral, and Granite, and include both dense and mixture-of-experts architectures.

These models also span a wide range of parameter scales. Among locally hosted models, `qwen3-32b` has the largest active-parameter count at 32.8B, while `nemotron-3-super` has the largest total-parameter count at 120B. In contrast, `granite-4-h-tiny` has the fewest active parameters at 1B, while `nemotron-3-nano-4b` is the smallest dense model at 4B total parameters. Although `gemma-4-e2b` is reported with 2.3B effective active parameters, its full checkpoint contains 5.1B total parameters due to per-layer embeddings and shared model components [15]. For API-accessible models, including `Claude-Sonnet-4.5`, `Gemini-3-Flash-Preview`, `GPT-5-mini`, and `o4-mini`, parameter counts have not been officially released.

Table 4: Overview of models evaluated on SprayBench, including release date and company, access type, active and total parameter counts (in Billions), and available input modalities.

Model	Year	Company	Access	Active Params (B)	Total Params (B)	Modality
`qwen3.5-35b-a3b`	2026	Alibaba	Open weights	3	35	Text-only
`qwen3.5-9b`	2026	Alibaba	Open weights	9	9	Text-only
`qwen3.6-27b`	2026	Alibaba	Open weights	27	27	Text-only
`qwen3.6-35b-a3b`	2026	Alibaba	Open weights	3	35	Text-only
`qwen3-30b-a3b`	2025	Alibaba	Open weights	3.3	30.5	Text-only
`qwen3-32b`	2025	Alibaba	Open weights	32.8	32.8	Text-only
`claude-sonnet-4-5`	2025	Anthropic	API	–	–	Multimodal
`gemma-4-12b`	2026	Google	Open weights	11.9	11.9	Multimodal
`gemma-4-12b-qat`	2026	Google	Open weights	11.9	11.9	Multimodal
`gemma-4-26b-a4b`	2026	Google	Open weights	3.8	25.2	Multimodal
`gemma-4-26b-a4b-qat`	2026	Google	Open weights	3.8	25.2	Multimodal
`gemma-4-31b`	2026	Google	Open weights	30.7	30.7	Multimodal
`gemma-4-31b-qat`	2026	Google	Open weights	30.7	30.7	Multimodal
`gemma-4-e2b`	2026	Google	Open weights	2.3	5.1	Multimodal
`gemma-4-e2b-qat`	2026	Google	Open weights	2.3	5.1	Multimodal
`gemma-4-e4b`	2026	Google	Open weights	4.5	8	Multimodal
`gemma-4-e4b-qat`	2026	Google	Open weights	4.5	8	Multimodal
`gemini-3-flash-preview`	2025	Google	API	–	–	Multimodal
`granite-4-h-tiny`	2025	IBM	Open weights	1	7	Text-only
`magistral-small`	2025	Mistral AI	Open weights	24	24	Text-only
`nemotron-3-nano-4b`	2026	NVIDIA	Open weights	4	4	Text-only
`nemotron-3-nano-omni`	2026	NVIDIA	Open weights	3.2	33	Multimodal
`nemotron-3-super`	2026	NVIDIA	Open weights	12	120	Text-only
`nemotron-3-nano`	2025	NVIDIA	Open weights	3.2	31.6	Text-only
`gpt-5-mini`	2025	OpenAI	API	–	–	Multimodal
`gpt-oss-120b`	2025	OpenAI	Open weights	5.1	117	Text-only
`gpt-oss-120b-high`	2025	OpenAI	Open weights	5.1	117	Text-only
`gpt-oss-120b-low`	2025	OpenAI	Open weights	5.1	117	Text-only
`gpt-oss-120b-medium`	2025	OpenAI	Open weights	5.1	117	Text-only
`gpt-oss-20b`	2025	OpenAI	Open weights	3.6	21	Text-only
`gpt-oss-20b-high`	2025	OpenAI	Open weights	3.6	21	Text-only
`gpt-oss-20b-low`	2025	OpenAI	Open weights	3.6	21	Text-only
`gpt-oss-20b-medium`	2025	OpenAI	Open weights	3.6	21	Text-only
`o4-mini`	2025	OpenAI	API	–	–	Multimodal

C Retrieval-Augmented Generation (RAG) Protocol

C.1 Corpus Construction and Chunking

The retrieval corpus was constructed from 5,205 articles on cold spray and additive manufacturing collected for SprayBench construction and evaluation. Each document was parsed into Markdown (`.md`) using MinerU [17] and segmented into overlapping windows of 5,000 characters with 500

characters of overlap, producing 80,161 unique chunks. Each chunk was assigned both a unique chunk identifier and a document identifier, allowing content filtering based on each question's source document.

C.2 Embedding, Re-ranking, and Context Selection

For each SprayBench question, we embedded the question using `BAAI/bge-m3` [4] and retrieved the top 40 candidate chunks by vector similarity. These candidates were then reranked with `BAAI/bge-reranker-v2-m3` [4]. After reranking, the top 8 eligible chunks were concatenated into the RAG context and inserted into the model prompt. The same retrieval budget, reranker, candidate chunk count, and final context size were used for both source-excluded and source-eligible RAG.

C.3 Document-Level Source Control

For each question, we recorded the source document used during benchmark construction. In source-excluded RAG, all chunks whose parent document matched the recorded source document were removed before final context selection. In source-eligible RAG, this exclusion was not applied. Thus, the original source document *could* be retrieved, but was not guaranteed to appear in the final top 8 context. In source-eligible RAG, similarity matching and chunk reranking identified the source document for 27.1% of matched questions. Source-excluded and source-eligible contexts were therefore identical for 72.9% of matched questions, as explored further in Appendix E.1.

D LLM-as-a-Judge Evaluation Protocol

As a benchmark for open-ended responses, deterministic scoring approaches such as exact match, multiple-choice parsing, or similarity matching are insufficient for SprayBench. We therefore evaluate each response with an LLM-as-a-judge protocol guided by the SME reference answer and scoring rubric. For each review, the judge receives the original question, the SME-refined target answer, and the candidate response. Our primary evaluation uses `GPT-5.4-mini`, but the protocol is model agnostic and can be implemented with any LLM. We also compare reviewer behavior against alternative models, including the larger `GPT-5.4`, the smaller `GPT-5.4-nano`, and several open-weight reviewers, `Qwen3.6-35b-a3b`, `GPT-OSS-120B`, `GPT-OSS-20B`, and `Gemma-4-E4B`.

D.1 Evaluation Metrics

In this work, each candidate response was evaluated using six numeric rubric criteria and a binary acceptability label. The numeric scores capture different aspects of technical answer quality, while the binary label determines whether the response passes the benchmark item.

The rubric criteria are:

- *Factual accuracy*: alignment with cold-spray science and the SME-refined target answer, including correct mechanisms, trends, and causal relationships.
- *Completeness*: coverage of the key concepts, caveats, tradeoffs, and process-structure-property links needed to answer the question.
- *Specificity*: technical precision, including material-specific, process-specific, or mechanism-specific detail rather than generic background discussion.
- *Relevance*: directness in answering the question, without drifting into unrelated cold spray or materials science content.
- *Absence of misconceptions*: avoidance of misleading claims, incorrect mechanisms, or unsupported process/property relationships.
- *Overall quality*: holistic judgment of whether the response is acceptable as a technical answer for evaluating cold spray domain knowledge.

Each criterion was scored on a 0 to 5 scale:

- **0**: no meaningful answer, a refusal, or a response dominated by severe technical errors.
- **1**: mostly incorrect or irrelevant, with major misconceptions or little usable domain content.
- **2**: partially relevant, but missing core mechanisms or key tradeoffs, or containing important errors.
- **3**: partially correct but incomplete, underspecified, or too generic for a strong technical answer.
- **4**: technically acceptable, with the main answer correct and only minor omissions or limitations.
- **5**: strong domain answer that is accurate, complete, specific, and well aligned with the SME-refined target.

The judge also returned a binary `acceptable` field used for pass rate reporting. This field records whether the candidate response would be acceptable for a cold spray domain knowledge evaluation. To ground this judgment, the prompt instructed the judge that an acceptable answer generally requires `overall_score` ≥ 4, `factual_accuracy` ≥ 4, and no `critical_errors`.

D.2 Judge Model Prompting Configurations

For LLM-as-a-judge prompting, we used a system prompt that defined the reviewer as a cold-spray and materials-science evaluator. The prompt instructed the reviewer to reward factual answer quality, technical specificity, relevant mechanisms, and alignment with the SME-refined target, while penalizing incorrect mechanisms, missing critical caveats, misleading claims, and unsupported process/property relationships. The prompt also instructed the reviewer not to reward verbosity, confident tone, formatting polish, or broad background discussion unless they improved factual answer quality. This was followed by a user prompt containing the rubric criteria defined in Appendix D.1, the question, the SME-refined target answer, the candidate response, and the required JSON output format. To improve the auditability of judge decisions beyond scalar scores, we also required the judge to outline claims made by the candidate model supported by the target answer, points that were missing from the candidate answer, critical errors, reviewer confidence, and a concise rationale for the acceptability decision. The complete system and user prompts are reported in Appendix F.1.

E Extended Results

E.1 Additional Source-Controlled RAG Results

E.1.1 Source-Excluded RAG: Question Rescue and Harm Analysis

In addition to measuring average pass rate changes from RAG, as shown in Table 1, we analyze paired pass/fail transitions between closed-book inference and source-excluded RAG. For this analysis, a response is *rescued* when it fails under closed-book inference but passes under source-excluded RAG, and *harmed* when it passes under closed-book inference but fails under source-excluded RAG.

As shown in Table 5, source-excluded RAG rescued more responses than it harmed across all matched configurations. Across $6{,}808$ matched responses, source-excluded RAG rescued $1{,}802$ responses and harmed 460, producing an average net gain of $+19.7$ pp over closed-book inference. The largest net gain occurred for `gpt-oss-20b-high`, where source-excluded RAG improved pass rate by $+29.4$ pp. In contrast, `gemma-4-26b-a4b` showed the highest harm rate, with harmful transitions accounting for 12.3% of all questions, although the model still improved by $+3.8$ pp overall.

Paired McNemar tests show that these gains are not simply due to stochastic pass/fail variation. Most evaluated configurations produced statistically significant rescue-harm imbalances at $p < 0.05$, while `gpt-5-mini` and `gemma-4-26b-a4b` were directionally positive but did not reach significance. Thus, while source-excluded RAG can introduce harmful context that degrades individual responses, it provides statistically significant aggregate gains at the dataset level, even when the question's source document is excluded.

Table 5: Source-excluded RAG rescue and harm analysis. Rows compare closed-book answers with source-excluded RAG on matched model-question responses. Rescued denotes fail-to-pass transitions; harmed denotes pass-to-fail transitions. McNemar tests compare paired closed-book and source-excluded RAG pass/fail decisions.

Model	No RAG	Source Excl.	Rescued	Harmed	Harm Rate	Net	McNemar p
`gpt-5-mini`	85.1%	88.2%	40	24	4.6%	+3.1 pp	0.0599
`qwen3.6-35b-a3b`	63.3%	73.2%	96	44	8.4%	+9.9 pp	1.3×10^{-5}
`gemma-4-26b-a4b`	57.3%	61.1%	84	64	12.3%	+3.8 pp	0.1180
`nemotron-3-super`	54.5%	71.9%	130	39	7.5%	+17.4 pp	1.2×10^{-12}
`gpt-oss-120b-medium`	40.5%	61.1%	147	39	7.4%	+20.6 pp	6.0×10^{-16}
`gpt-oss-120b-low`	39.9%	60.1%	155	49	9.4%	+20.2 pp	5.1×10^{-14}
`gpt-oss-120b`	38.0%	62.2%	160	33	6.3%	+24.2 pp	3.3×10^{-21}
`gpt-oss-120b-high`	37.4%	64.1%	176	36	6.9%	+26.7 pp	2.4×10^{-23}
`gemma-4-e4b`	24.8%	49.8%	155	24	4.6%	+25.0 pp	1.2×10^{-24}
`gpt-oss-20b-low`	23.9%	46.9%	156	35	6.7%	+23.1 pp	2.0×10^{-19}
`gpt-oss-20b`	22.5%	47.9%	159	26	5.0%	+25.4 pp	1.7×10^{-24}
`gpt-oss-20b-high`	21.0%	50.4%	176	22	4.2%	+29.4 pp	5.1×10^{-31}
`gpt-oss-20b-medium`	20.4%	47.7%	168	25	4.8%	+27.3 pp	3.3×10^{-27}
Total	40.6%	60.4%	1,802	460	6.8%	+19.7 pp	8.7×10^{-187}

E.1.2 Source-Eligible RAG: Identifying Source-Retrieval Gains

In addition to the rescue-harm analysis, we separate RAG gains into broad literature-grounding gains and gains associated with direct answer-source retrieval. We define broad grounding gain (BG) as source-excluded RAG minus closed-book performance, and source eligibility gain as source-eligible RAG minus source-excluded RAG. Source-eligible RAG retrieved the originating source document for 27.1% of matched questions. We therefore separate the source eligibility gain into *hit* questions, where the source document appeared in the final top 8 context, and *miss* questions, where it did not. As shown in Table 6, miss questions showed little change in performance (-0.2 pp), while hit questions gained $+14.5$ pp over source-excluded RAG. Across 6,808 matched rows, source-excluded RAG accounted for most of the total retrieval benefit, improving pass rate by $+19.7$ pp over closed-book inference, while source eligibility added $+3.8$ pp.

Table 6: RAG pass-rate gain partitioning by answer-source retrieval. BG reports the gain from source-excluded RAG over closed-book inference, while ITT reports the additional gain from source-eligible RAG over source-excluded RAG. Hit and Non-Hit separate the ITT contrast by whether the originating source document appeared in the final top-8 context. Wald reports ITT divided by the source-hit rate as a descriptive exposure-opportunity summary, and Total reports source-eligible RAG minus closed-book performance.

Cohort	No RAG	Excl.	Incl.	BG	ITT	Hit	Non-Hit	Wald	Total
`gpt-5-mini`	85.1%	88.2%	91.4%	+3.1 pp	+3.2 pp	+11.2 pp	+0.3 pp	+11.9 pp	+6.3 pp
`qwen3.6-35b-a3b`	63.3%	73.2%	77.2%	+9.9 pp	+4.0 pp	+20.3 pp	−2.1 pp	+14.7 pp	+14.0 pp
`gemma-4-26b-a4b`	57.3%	61.1%	65.7%	+3.8 pp	+4.6 pp	+16.8 pp	+0.0 pp	+16.8 pp	+8.4 pp
`nemotron-3-super`	54.5%	71.9%	74.8%	+17.4 pp	+2.9 pp	+12.6 pp	−0.8 pp	+10.5 pp	+20.3 pp
`gpt-oss-120b-medium`	40.5%	61.1%	64.5%	+20.6 pp	+3.4 pp	+10.5 pp	+0.8 pp	+12.6 pp	+24.0 pp
`gpt-oss-120b-low`	39.9%	60.1%	66.8%	+20.2 pp	+6.7 pp	+16.8 pp	+2.9 pp	+24.5 pp	+26.9 pp
`gpt-oss-120b`	38.0%	62.2%	68.3%	+24.2 pp	+6.1 pp	+16.1 pp	+2.4 pp	+22.4 pp	+30.3 pp
`gpt-oss-120b-high`	37.4%	64.1%	68.7%	+26.7 pp	+4.6 pp	+16.8 pp	+0.0 pp	+16.8 pp	+31.3 pp
`gemma-4-e4b`	24.8%	49.8%	51.9%	+25.0 pp	+2.1 pp	+7.7 pp	+0.0 pp	+7.7 pp	+27.1 pp
`gpt-oss-20b-low`	23.9%	46.9%	50.0%	+23.1 pp	+3.1 pp	+11.9 pp	−0.3 pp	+11.2 pp	+26.1 pp
`gpt-oss-20b`	22.5%	47.9%	51.0%	+25.4 pp	+3.1 pp	+13.3 pp	−0.8 pp	+11.2 pp	+28.4 pp
`gpt-oss-20b-high`	21.0%	50.4%	51.3%	+29.4 pp	+1.0 pp	+13.3 pp	−3.7 pp	+3.5 pp	+30.3 pp
`gpt-oss-20b-medium`	20.4%	47.7%	52.7%	+27.3 pp	+5.0 pp	+21.0 pp	−1.0 pp	+18.2 pp	+32.3 pp
All matched rows	40.6%	60.4%	64.2%	+19.7 pp	+3.8 pp	+14.5 pp	−0.2 pp	+14.0 pp	+23.5 pp

E.2 Judge Reliability and Calibration

SprayBench questions are open-ended and evaluated across many model, retrieval, and reviewer configurations, making deterministic scoring insufficient and exhaustive SME review impractical. We therefore use an LLM-as-a-judge protocol to evaluate candidate responses at scale. This provides a practical approach to grading technical answers, but also introduces additional sources of uncertainty, including judge model bias, stochastic review noise, reviewer drift, decision boundary sensitivity, and limited domain knowledge from the reviewer. To reduce and measure these risks, our primary evaluator is `GPT-5.4-mini`, each response receives three independent reviews, and reviewer behavior is compared with the larger `GPT-5.4` reviewer and several open-weight alternatives. This allows us to evaluate candidate model performance, in addition to the reliability and calibration of the scoring protocol used to measure it.

E.2.1 Repeated Review Reliability

To evaluate stochastic review noise, we repeated judge evaluation on model-question responses and measured reliability under different review budgets. Reliability was measured using *intraclass correlation coefficients* (ICCs) [9], where higher values indicate stronger agreement across repeated reviews of the same response. As shown in Table 7, single-review reliability was already strong, with an ICC of 0.912 for `GPT-5.4-mini` and 0.935 for `GPT-5.4`. Averaging repeated reviews further increased ICC to 0.954 and 0.968 for two and three `GPT-5.4-mini` reviews, respectively, and to 0.966 and 0.977 for `GPT-5.4`.

Table 7: Reliability of repeated judge reviews under different review budgets. We report two-way random-effects absolute-agreement ICCs: $\mathrm{ICC}(A,1)$ for one review and $\mathrm{ICC}(A,k)$ for the mean of k repeated reviews of the same model-question response.

Judge Model	Review Budget	N	Overall ICC	Δ Overall ICC
`GPT-5.4-Mini`	1 review	20,112	0.912 [0.91, 0.91]	–
`GPT-5.4-Mini`	Mean of 2 reviews	20,112	0.954 [0.95, 0.96]	+0.042
`GPT-5.4-Mini`	Mean of 3 reviews	20,112	0.968 [0.97, 0.97]	+0.014
`GPT-5.4` (Full)	1 review	20,112	0.935 [0.93, 0.94]	–
`GPT-5.4` (Full)	Mean of 2 reviews	20,112	0.966 [0.97, 0.97]	+0.031
`GPT-5.4` (Full)	Mean of 3 reviews	20,112	0.977 [0.98, 0.98]	+0.011

These results demonstrate that `GPT-5.4-mini` and `GPT-5.4` have strong score reliability within each judge model, but individual pass/fail reviews can still vary across runs, motivating multiple reviews. For example, of the 19,588 eligible `GPT-5.4-mini` reviewed rows with confidence labels, 2,908 (14.8%) produced split pass/fail votes across the three reviews. Disagreement across repeated reviews from the same judge model was also strongly associated with reported reviewer confidence. Responses with at least one medium confidence review had a 33.5% split vote rate, compared with 7.8% when all three reviews were high confidence. First review confidence also provided an early uncertainty signal, with medium confidence first reviews producing a 32.9% split vote rate compared with 12.3% for high confidence first reviews. As a result, reviewer uncertainty can serve as one criterion for adaptive evaluation, as explored further in Appendix E.4.

E.2.2 GPT-5.4-Mini vs. GPT-5.4: Model Alignment, Drift and Decision Boundary Effects

Repeated reviews improve reliability within a judge model, but do not guarantee alignment between different judge models. To measure calibration between judge models, we treated `GPT-5.4-mini` and `GPT-5.4` as separate reviewers on matched model-question responses. This produced an ICC of 0.874 for *Overall Score*, indicating strong alignment between the lower cost `GPT-5.4-mini` and the larger `GPT-5.4` for continuous score evaluation. However, binary pass/fail decisions, using majority vote, showed more disagreement. Across 20,112 matched responses, the two judges disagreed on 2,468 rows, a 12.3% flip rate. As shown in Table 8, rather than appearing as uniform stochastic noise, disagreements were concentrated at the decision boundary. All 2,468 disagreements occurred on rows assigned rounded mean `GPT-5.4` scores from 2 to 4, and 2,188 (88.7%) occurred at score 3.

In addition to being localized at the decision boundary, as shown in Table 8, mini-full disagreement was also frequently directional. Of the 2,468 disagreements, 2,250 (91.2%) occurred when `GPT-5.4`

Table 8: Decision-boundary disagreement between `GPT-5.4-Mini` and `GPT-5.4`. Rows group matched model-question responses by the rounded mean `GPT-5.4` overall score. Pass/fail decisions use each reviewer family's majority vote across repeated reviews. Disagreements primarily occur at score 3, where `GPT-5.4-Mini` frequently accepts responses that `GPT-5.4` rejects. Row flip rate is the within-score disagreement rate, and total flip share is the fraction of all matched responses represented by that row's disagreements.

`GPT-5.4` Score	N	Disagreement	Mini+/Full-	Mini-/Full+	Row Flip Rate	Total Flip Share
0	398	0	0	0	0.0%	0.0%
1	2,475	0	0	0	0.0%	0.0%
2	5,341	60	60	0	1.1%	0.3%
3	6,330	2,188	2,188	0	34.6%	10.9%
4	4,833	220	2	218	4.6%	1.1%
5	735	0	0	0	0.0%	0.0%
Total	20,112	2,468	2,250	218	12.3%	12.3%

rejected a response that `GPT-5.4-mini` accepted, with 2,188 of these occurring when `GPT-5.4` assigned the response a score of 3. Similar to repeated review disagreement, `GPT-5.4-mini` uncertainty was strongly associated with mini/full disagreement: 28.1% of rows showed disagreement when any mini review reported medium confidence, compared with 6.5% when all three mini reviews reported high confidence. Thus, reviewer confidence, boundary localization, and disagreement direction provide practical signals for adaptive review, as explored further in Appendix E.4. Additionally, despite this response level disagreement, `GPT-5.4-mini` preserved strong model level agreement with `GPT-5.4`. Average overall score rankings retained a strong alignment (Spearman $\rho = 0.994$, Kendall $\tau_b = 0.950$), with a mean absolute rank displacement of 0.7 positions.

Finally, these results show overall directional drift, with `GPT-5.4-mini` more lenient in overall score and pass/fail acceptability, but this drift was not constant across rubric criteria. As shown in Table 9, `GPT-5.4-mini` was more lenient on *Factual Accuracy* and *Absence of Misconceptions*, with mean shifts of $+0.190$ and $+0.276$, respectively. In contrast, `GPT-5.4-mini` was stricter on *Completeness* and *Relevance*, with mean shifts of -0.096 and -0.294, while *Specificity* was nearly symmetric (-0.014). These results further demonstrate that candidate models *and* reviewer models should be evaluated, since reviewer drift is not constant across rubric criteria and can affect different evaluation conclusions.

Table 9: Mini-full rubric asymmetry by scoring dimension. Mean deltas are mini minus full on matched rows; positive values indicate mini leniency. Directional percentages report how often one evaluator scored above the other on the same row.

Rubric	Mean Δ	Mean $\lvert\Delta\rvert$	Pearson r	Spearman ρ	Mini > Full	Full > Mini	Identical	Direction
Factual Accuracy	$+0.190$	0.346	0.878	0.871	25.9%	7.7%	66.4%	Mini Lenient
Completeness	-0.096	0.351	0.888	0.882	12.5%	21.8%	65.7%	Mini Stricter
Specificity	-0.014	0.372	0.872	0.861	17.2%	18.8%	64.0%	Symmetric
Relevance	-0.294	0.423	0.783	0.730	6.3%	29.9%	63.8%	Mini Stricter
Absence of Misconceptions	$+0.276$	0.511	0.861	0.867	32.7%	11.0%	56.3%	Mini Lenient
Overall Continuous Score	$+0.130$	0.346	0.881	0.876	23.1%	10.7%	66.2%	Mini Lenient

E.2.3 Additional Judge Model Alternatives

These results show that `GPT-5.4-mini` reduces reviewer cost while maintaining strong model-level alignment and rank preservation with `GPT-5.4`. However, depending on the number of candidate models, inference configurations, and repeated reviews, `GPT-5.4-mini` may still be cost-prohibitive. Thus, we also evaluated `GPT-5.4-nano`, a smaller reviewer in the same model family with lower token costs ($0.20/M input tokens, $1.25/M output tokens). However, while enabling lower inference costs, all `GPT-5.4`-family reviewers require API access, which is not compatible with fully local, private, or information-regulated materials workflows. To address this, we also evaluated four open-weight reviewers that can be run on the same DGX Spark hardware used throughout this study: `Qwen3.6-35B-A3B`, `GPT-OSS-120B`, `GPT-OSS-20B`, and `Gemma-4-E4B`.

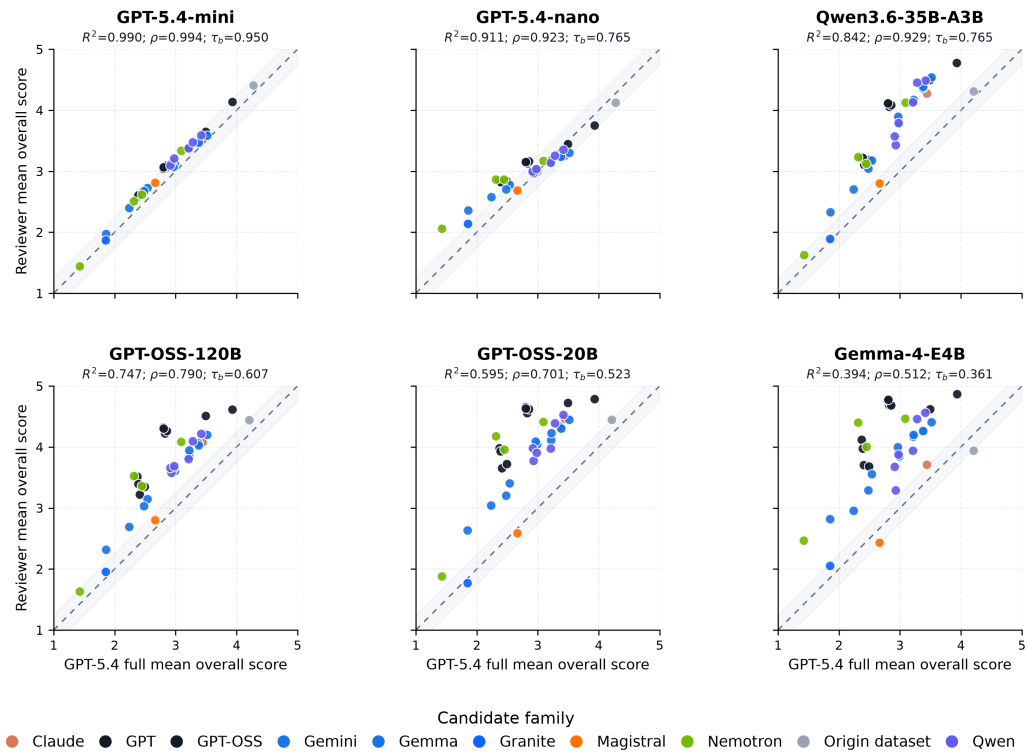


Figure 2: **Judge-model alignment with `GPT-5.4`.** Model-level mean overall scores from alternative reviewer models compared with `GPT-5.4` on matched candidate-model responses. Each point represents one candidate model configuration, colored by candidate family. The dashed line marks perfect agreement with `GPT-5.4`, while the gray band indicates standard deviation across repeated `GPT-5.4` reviews.

These open-weight alternatives were selected from the 34 primary candidate models tested to span models that performed well, such as `Qwen3.6-35B-A3B`, models with high total parameter counts, such as `GPT-OSS-120B`, and smaller or lower-performing models, such as `GPT-OSS-20B` and `Gemma-4-E4B`. This selection provides insight into how underlying model capability affects alignment with API-based reviewers such as `GPT-5.4` and `GPT-5.4-mini`. For reviewer inference, `GPT-5.4-nano` was evaluated with the same three-review protocol used for the other `GPT-5.4` reviewers. In contrast, open-weight reviewers were run once per response with temperature `0.0`, using the same local inference settings described in Appendix B. All reviewer models used the same rubric-based judge protocol described in Appendix D.2, enabling comparison of score-scale alignment, pass/fail agreement, and model-level rank preservation across reviewer families.

As shown in Figure 2, reviewer drift generally increased as reviewer capability decreased during continuous overall-score evaluation. `GPT-5.4-mini` aligned most closely with `GPT-5.4` ($R^2 = 0.990$, Spearman $\rho = 0.994$, Kendall $\tau_b = 0.950$), whereas `Gemma-4-E4B` exhibited the highest drift ($R^2 = 0.394$, $\rho = 0.512$, $\tau_b = 0.361$). Within the `GPT-5.4` family, `GPT-5.4-nano` retained relatively strong score alignment ($R^2 = 0.911$, $\rho = 0.923$, $\tau_b = 0.765$), though not as strong as `GPT-5.4-mini`. `Qwen3.6-35B-A3B` retained high rank correlation ($R^2 = 0.842$, $\rho = 0.929$, $\tau_b = 0.765$), but showed larger score-scale drift than `GPT-5.4-nano`, especially for top-performing candidate models. In contrast, `GPT-5.4-nano` remained closer to the `GPT-5.4` score scale for these models and was occasionally more conservative than `GPT-5.4`.

Compared with continuous overall scores, binary pass-fail decisions showed stronger reviewer-leniency effects and weaker agreement with `GPT-5.4`, reducing model-level R^2 for reviewers, including `GPT-5.4-mini` ($0.990 \rightarrow 0.968$), `GPT-5.4-nano` ($0.911 \rightarrow 0.866$), and `Qwen3.6-35B-A3B` ($0.842 \rightarrow 0.737$). This is consistent with the results above, where small reviewer differences near the decision boundary can amplify model-level pass-rate disagreement. However, despite stronger

leniency and lower R^2, pass-fail decisions often preserved similar or slightly higher model-level Spearman correlation and Kendall rank agreement, indicating that reviewer drift affected absolute pass rates more than coarse model ordering.

E.2.4 Human-Reviewer Audit

To evaluate agreement between LLM reviewers and human technical judgment, a Ph.D.-level cold-spray expert reviewed a 105-response subset of closed-book inference outputs. The audit sampled unique model-question response pairs, including three responses per inference model, with at least one response receiving a majority `Pass` from `GPT-5.4-mini` and at least one receiving a majority `Fail`. Sampling also limited repeated use of the same question across inference models and enforced level diversity, producing 35 responses each from Level 1, Level 2, and Level 3. The final 105-response audit was blinded for human review, with model identity, question level, and Mini-judge decisions withheld.

Comparing human reviews to LLM-as-a-judge reviews, we found that although `GPT-5.4` was stricter than `GPT-5.4-mini`, both reviewers enforced a stricter pass-fail threshold than the human reviewer. On the 105-response audit subset, the human reviewer rated 86 responses as acceptable, compared with 52 accepted by `GPT-5.4-mini` and 38 accepted by `GPT-5.4`. While this shows only a moderate pass-fail alignment, disagreement was highly directional. Namely, `GPT-5.4` produced no human-fail/LLM-pass cases and 48 human-pass/LLM-fail cases, while `GPT-5.4-mini` produced two human-fail/LLM-pass cases and 36 human-pass/LLM-fail cases.

Although binary pass-fail agreement was limited, continuous overall scores retained clearer alignment with human scoring. For overall score, `GPT-5.4` reached a Pearson correlation of 0.616 with human scores, while `GPT-5.4-mini` reached 0.629. However, both reviewers scored responses lower than the human reviewer on average, with mean overall scores of 3.05 and 3.18 compared with the human mean of 4.11. Feature-level differences show that this gap was driven mainly by stricter penalties for completeness, factual accuracy, and absence of misconceptions. In contrast, both LLM reviewers scored relevance higher than the human reviewer. These results highlight that, in this audit, LLM-as-a-judge evaluation applied stricter acceptability thresholds than human review, with strongly directional disagreements driven primarily by LLM rejection of human-accepted responses.

E.2.5 Impact of SME-Refinement on Candidate Model Evaluation Performance

In this work, we used a two-stage human-in-the-loop pipeline to create SprayBench. In Stage 1, `o4-mini` automatically generated source-anchored question-answer pairs from scientific literature. In Stage 2, these answers were refined by a cold-spray subject-matter expert, correcting factual errors, adding missing context, and aligning answers toward practitioner-facing explanation, tradeoff analysis, and process guidance. As a result of SME refinement, target answers increased from 91.4 to 141.1 words on average, while retaining 46.6% of the original auto-extracted tokens. When evaluated against the SME-refined targets using the same LLM-as-a-judge pipeline used for candidate-model scoring, the original auto-extracted answers achieved a 90.5% pass rate, outperforming every closed-book candidate model tested in this work.

While SME refinement improves target quality, auditability, and practitioner alignment, it also requires substantial expert effort. To measure how much this refinement affects model-level evaluation conclusions, we performed a secondary analysis using the same LLM-as-a-judge pipeline, but treating the auto-extracted answers as the target answers. We then compared model performance under auto-extracted targets against performance under SME-refined targets.

As shown in Figure 3, auto-extracted targets produced evaluation results that were strongly aligned with SME-refined targets at the model level. For reported accept rate, auto-extracted and SME-refined evaluations showed strong agreement ($R^2 = 0.981,\ \rho = 0.985,\ \tau_b = 0.917$), with overall scores showing even stronger agreement ($R^2 = 0.996,\ \rho = 0.993,\ \tau_b = 0.946$). However, similar to the reviewer-model drift observed in Appendix E.2.3, auto-extracted targets were generally more lenient than SME-refined targets, producing higher absolute pass rates and scores for many candidate models.

Ultimately, these results highlight a tradeoff in benchmark construction. SME refinement adds technical context and refines target answers, providing a more rigorous evaluation of model performance. However, strong agreement between auto-extracted and SME-refined evaluations suggests that stage-one extraction can still recover broad model-level performance trends on SprayBench.

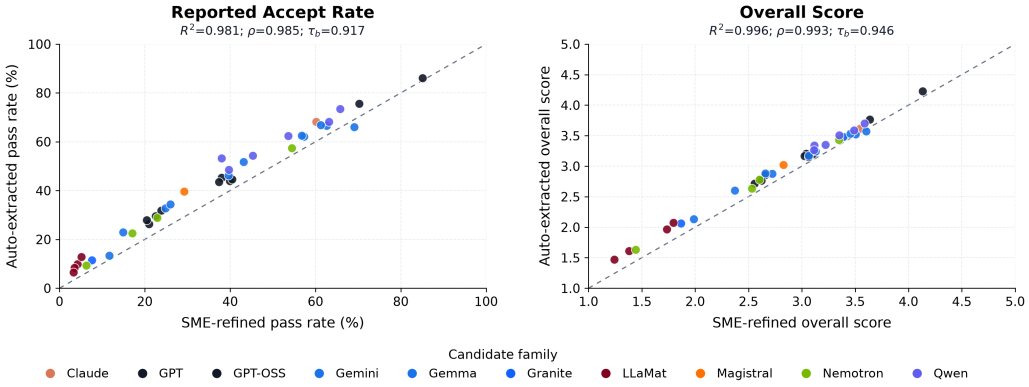


Figure 3: **Effect of SME-refined versus auto-extracted judge targets.** Model-level SprayBench performance using auto-extracted targets compared with SME-refined targets with `GPT-5.4-mini`. Each point represents one candidate model configuration, colored by candidate family. The dashed line marks perfect agreement between target-answer conditions. Auto-extracted targets preserve model-level rankings but produce more lenient absolute accept rates and overall scores.

Thus, SME-refined targets remain necessary for final scoring, while auto-extracted targets may provide a useful proxy for preliminary model screening or larger-scale benchmark expansion when expert availability is prohibitive.

E.3 Model-Factor Analysis

In this work, we evaluate a diverse set of open-weight LLMs spanning dense and mixture-of-experts architectures, small models with as few as 4 billion total parameters, and large models with up to 117 billion total parameters. These models also come from different model families and release generations, with distinct pretraining, post-training, reasoning, and deployment characteristics. Thus, in addition to reporting per-model SprayBench performance, we performed a model-factor analysis to examine which model-level properties were associated with stronger open-ended cold-spray reasoning.

E.3.1 Model Factors and Covariance

For this analysis, we evaluated six model factors: *total parameters*, *active parameters*, *output verbosity*, defined as the number of tokens in the final response, *reasoning verbosity*, defined as total generated tokens minus output tokens; *input token length*, defined as the number of tokens in the question, and a binary *architecture* variable, set to 0 for dense models and 1 for MoE architectures. We restricted this analysis to the 30 open-weight models used in our primary evaluation, excluding API-based models because their exact parameter counts, active parameter counts, and architectural configurations are not consistently available.

However, because many of these factors are correlated to each other, such as total and active parameter counts in dense models, we measured pairwise correlations using Spearman rank correlation. As shown in Table 10, only three pairwise relationships were statistically significant after Benjamini-Hochberg correction. MoE architecture was positively associated with total parameter count ($\rho = +0.53$), negatively associated with active parameter count ($\rho = -0.47$), and positively associated with input token length ($\rho = +0.61$).

E.3.2 Model Factors and SprayBench Performance

In addition to inter-parameter correlation, we evaluated the correlation between model factors and SprayBench performance, as shown in Table 11. Using Benjamini-Hochberg FDR-adjusted q-values, total parameter count, active parameter count, and reasoning verbosity remained the factors most consistently associated with SprayBench performance. While prior work has shown that LLM-as-

Table 10: Spearman correlation among model factors, including total parameters, active parameters, reasoning verbosity, output verbosity, input token length, and a binary indicating dense or mixture architecture. Lower-triangle cells show Spearman ρ, while stars mark Benjamini-Hochberg FDR-adjusted q-values: $*q < 0.05$, $**q < 0.01$, $***q < 0.001$, and values are bolded when $q < 0.05$.

Factor	Total Params	Active Params	Reasoning Verbosity	Output Verbosity	Input Token Length	MoE Arch
Total parameters	$+1.00$	—	—	—	—	—
Active parameters	$+0.27$	$+1.00$	—	—	—	—
Reasoning verbosity	$+0.17$	$+0.16$	$+1.00$	—	—	—
Output verbosity	$+0.42$	$+0.20$	-0.16	$+1.00$	—	—
Input token length	$+0.43$	-0.11	-0.17	$+0.38$	$+1.00$	—
MoE architecture	$\mathbf{+0.53}^{*}$	$\mathbf{-0.47}^{*}$	-0.12	$+0.30$	$\mathbf{+0.61}^{**}$	$+1.00$

a-judge evaluation can be biased toward longer or overly confident outputs [21], we observed no statistically significant correlation between output verbosity and SprayBench pass rate or overall score. Output verbosity was significantly associated with Completeness ($\rho = +0.42$), but was not significantly associated with pass rate, overall score, or most other rubric dimensions. In contrast, reasoning verbosity remained significantly correlated with every SprayBench performance metric, indicating that more expansive model reasoning capabilities were associated with more accurate, relevant, and misconception-free technical answers.

Table 11: Spearman correlation between model factors, including total parameters, active parameters, reasoning verbosity, output verbosity, input token length, and a binary indicating dense or mixture architecture with model performance. Each cell shows Spearman ρ, while stars mark Benjamini-Hochberg FDR-adjusted q-values: $*q < 0.05$, $**q < 0.01$, $***q < 0.001$, and values are bolded when $q < 0.05$.

Factor	Pass rate	Overall	Factual	Complete	Specific	Relevant	No misconcep.
Total parameters	$\mathbf{+0.53}^{**}$	$\mathbf{+0.51}^{*}$	$\mathbf{+0.52}^{**}$	$\mathbf{+0.76}^{***}$	$\mathbf{+0.66}^{***}$	$\mathbf{+0.54}^{**}$	$\mathbf{+0.47}^{*}$
Active parameters	$\mathbf{+0.55}^{**}$	$\mathbf{+0.54}^{**}$	$\mathbf{+0.52}^{**}$	$\mathbf{+0.50}^{*}$	$\mathbf{+0.46}^{*}$	$\mathbf{+0.55}^{**}$	$\mathbf{+0.54}^{**}$
Reasoning verbosity	$\mathbf{+0.64}^{***}$	$\mathbf{+0.68}^{***}$	$\mathbf{+0.70}^{***}$	$\mathbf{+0.50}^{*}$	$\mathbf{+0.58}^{**}$	$\mathbf{+0.68}^{***}$	$\mathbf{+0.68}^{***}$
Output verbosity	$+0.09$	$+0.05$	$+0.02$	$\mathbf{+0.42}^{*}$	$+0.33$	-0.01	-0.04
Input token length	-0.17	-0.19	-0.19	$+0.18$	$+0.11$	-0.16	-0.26
MoE architecture	-0.04	-0.05	-0.05	$+0.19$	$+0.18$	-0.03	-0.10

Similarly, both total and active parameter count showed statistically significant correlation with SprayBench performance over all performance metrics, indicating that, in this analysis, larger models generally performed better. Interestingly, total parameter count was most strongly associated with *Completeness* and *Specificity*, while active parameter count was more evenly associated across performance metrics. This aligns with expected dense and MoE behavior, where an increase in active parameters is strongly correlated with a model's ability to produce high-quality answers, while a larger total parameter count enables the model to retain more breadth and technical detail during training.

E.3.3 Verbosity, Token Length, and Score Variance

Building on these results, we further evaluated the impact of input token length, reasoning verbosity, and output verbosity on LLM-as-a-judge evaluations across all 34 candidate models, including the four API-based models. Output verbosity was positively but not significantly correlated with SprayBench pass rate or overall score after correction, while showing a small but statistically significant association with judge-score variance. Specifically, using a GEE model on log repeated-review score variance clustered by question, a one-standard-deviation increase in output verbosity was associated with higher residual score variance ($\beta = +0.023, p = 1.8 \times 10^{-6}$). Input token length showed a similar variance effect ($\beta = +0.024, p = 2.2 \times 10^{-6}$) and was negatively associated with pass rate and overall score, indicating that longer questions or contexts were both harder for candidate models and more variable for reviewers. In contrast, reasoning verbosity remained significantly correlated with stronger SprayBench performance, but was not associated with increased score variance ($\beta = +0.006, p = 0.1687$). Thus, longer final answers and longer inputs were associated with higher evaluation heteroscedasticity, while reasoning verbosity was more directly associated with stronger performance without increasing reviewer variance.

Table 12: Verbosity and token-length associations with performance and score heteroscedasticity. Spearman ρ columns report model-level associations with pass rate and overall score; q-values are Benjamini-Hochberg adjusted. GEE log-variance β estimates whether each standardized predictor changes residual score variance after clustering by item prompt.

Predictor	ρ_{pass}	q_{pass}	ρ_{score}	q_{score}	GEE Log-Var β	Variance p-value
Output Verbosity	$+0.34$	0.0597	$+0.31$	0.0822	$+0.023$	1.8×10^{-6}
Reasoning Verbosity	$+0.68$	3.5×10^{-5}	$+0.68$	3.5×10^{-5}	$+0.006$	0.1687
Input Token Length	-0.47	0.0071	-0.49	0.0053	$+0.024$	2.2×10^{-6}

E.4 SprayRouter: Adaptive Judge Model Selection

Motivation. In the prior results, we demonstrated strong alignment between `GPT-5.4` and smaller reviewers in the same model family, `GPT-5.4-mini` and `GPT-5.4-nano`, at lower token costs. At the time of writing, `GPT-5.4` costs \$2.50/M input tokens and \$15.00/M output tokens, compared with \$0.75/M and \$4.50/M for `GPT-5.4-mini`, and \$0.20/M and \$1.25/M for `GPT-5.4-nano`. However, this analysis assumed a fixed reviewer, where every candidate response is evaluated by the same judge model. In our evaluator-calibration results, reported in Section 3 and Appendix E.2.2, we observed that reviewer disagreement was non-uniform, concentrated near the pass/fail boundary, and occurred more frequently when reviewer models reported lower confidence. As a result, we present **SprayRouter**, a hierarchical reviewer-routing strategy that uses cheaper reviewers for low-risk cases and escalates uncertain responses to stronger reviewers, preserving high agreement with the `GPT-5.4` reference at lower cost.

Routing setup. We evaluated SprayRouter on 17,504 matched model-question rows across the 34 primary candidate models, treating the three-review `GPT-5.4` majority consensus as the reference target. We compared fixed single-reviewer baselines against two adaptive routing strategies: a mini-first router and a nano+mini router. For cheaper reviewer stages, single `GPT-5.4-nano` and `GPT-5.4-mini` reviews were first translated into pseudo-`GPT-5.4` single-reviewer judgments using out-of-fold ridge regression models [8], trained to predict single-reviewer outputs from reviewer scores, acceptability, response length, token counts, and review cost. Additionally, we trained logistic escalation models to estimate the probability that the currently available decision, either nano pseudo-full, mini pseudo-full, or the first `GPT-5.4` decision, depending on the stage, would disagree with a three-review `GPT-5.4` consensus. For the mini-first router, this produced two model-driven escalation stages: one after the initial `GPT-5.4-mini` review and one after the first `GPT-5.4` review. If the first `GPT-5.4` review was predicted to be high risk, the router "*purchased*" a second `GPT-5.4` review, while a third `GPT-5.4` review was used only when the first two `GPT-5.4` reviews disagreed. The nano+mini router followed the same escalation logic, but added an initial `GPT-5.4-nano` review and escalation stage prior to `GPT-5.4-mini`.

Threshold selection. Using the logistic escalation models, we tested several calibration thresholds, enabling the exploration of cost-performance tradeoffs in dynamic reviewer selection. To reduce potential overfitting, the reviewer translators and escalation models were evaluated in a leave-one-family-out setting, where, for each candidate-model family (e.g., `Qwen`, `Gemma`, `Nemotron`, etc.), the router was trained on all other families and then used to predict escalation risk for the held-out family. After repeating this across all families, we pooled the held-out risk predictions and swept escalation thresholds over the combined cohort. These thresholds were selected under four configurations: matching single-`GPT-5.4` agreement (96.9%) at minimum cost, targeting a higher 98% agreement, maximizing agreement at or below the cost of one `GPT-5.4` review, and targeting 99% agreement even when additional review cost was required.

Agreement-cost tradeoff. As shown in Table 13, using a single fixed judge model reduced reviewer cost at varying levels of agreement. For example, a single `GPT-5.4` review reduced cost to 33% of the three-review reference while retaining 96.9% agreement. In comparison, single `GPT-5.4-mini` and `GPT-5.4-nano` reviews reduced cost further to 12% and 2%, respectively, but reached only 86.5% and 84.4% agreement. This suggests that cheaper reviewers are useful for triage, but not as direct replacements for `GPT-5.4`. Across the adaptive settings, *nano+mini* routing frequently achieved similar agreement to *mini-first* routing at lower cost. When matching single-`GPT-5.4` agreement, nano+mini reached 97.0% agreement at $0.78\times$ the cost of one `GPT-5.4` review, compared with 97.0% agreement at $0.84\times$ cost for mini-first. At the balanced 98% setting, nano+mini reached 98.1%

agreement at 0.93× the cost of one `GPT-5.4` review. Under a comparable one-`GPT-5.4`-review budget, nano+mini produced a stronger operating point, reaching 98.4% agreement with 273 errors. At the strictest threshold tested, targeting 99% agreement, mini-first achieved 99.1% agreement at 39% of the three-review `GPT-5.4` reference cost, while nano+mini achieved 99.0% agreement at 38% of the reference cost.

Table 13: **SprayRouter operating points.** Agreement and cost tradeoffs for fixed and adaptive reviewer strategies, evaluated against the three-review `GPT-5.4` consensus. Cost is reported relative to one or three `GPT-5.4` reviews.

Setting	Strategy	Agreement	Cost / 1× `GPT-5.4`	Cost / 3× `GPT-5.4`	Errors
Fixed baseline	1× `GPT-5.4-nano`, translated	84.4%	0.06	0.02	2,726
Fixed baseline	1× `GPT-5.4-mini`, translated	86.5%	0.37	0.12	2,355
Fixed baseline	1× `GPT-5.4`	96.9%	1.00	0.33	536
Match 1× `GPT-5.4`	Mini-first	97.0%	0.84	0.28	521
Match 1× `GPT-5.4`	Nano+mini	97.0%	0.78	0.26	531
Balanced 98%	Mini-first	98.0%	0.96	0.32	347
Balanced 98%	Nano+mini	98.1%	0.93	0.31	330
Best ≤ 1× `GPT-5.4`	Mini-first	98.2%	0.99	0.33	313
Best ≤ 1× `GPT-5.4`	Nano+mini	98.4%	1.00	0.33	273
High-agreement 99%	Mini-first	99.1%	1.17	0.39	163
High-agreement 99%	Nano+mini	99.0%	1.13	0.38	175

Held-out validation. Building on Table 13, which reports operating points selected from pooled family-out-of-fold risk scores, we evaluated three nested held-out validation settings. In these experiments, escalation thresholds were selected within each training split and then evaluated on held-out questions (10 folds), candidate models (34 folds), or model families (10 folds). As shown in Figure 4, nano+mini routing preserved the same general cost-performance tradeoff observed in the pooled analysis, achieving comparable agreement to mini-first while often retaining greater cost reduction. The leave-question-out setting provides an additional check that SprayRouter is not simply learning which benchmark questions tend to require escalation, since held-out questions were excluded from router training and threshold selection before being evaluated as held-out test folds.

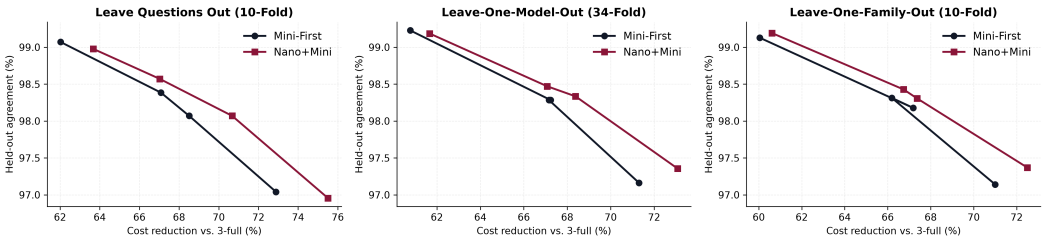


Figure 4: **Comparing SprayRouter configurations on held-out validation sets.** Held-out agreement and cost reduction for mini-first and nano+mini routing under leave-question-out, leave-model-out, and leave-family-out validation. Agreement is measured against the three-review `GPT-5.4` consensus, and cost reduction is reported relative to evaluating every response with three `GPT-5.4` reviews. Each point represents one routing policy selected on training folds to satisfy a target agreement level while minimizing review cost, as described in Table 13.

Key Takeaways. These results highlight that cheaper reviewers can reduce evaluation cost when used adaptively, but cannot serve as direct replacements for a calibrated stronger judge model. A single cheaper reviewer can reduce cost while preserving strong agreement in low-risk cases, but the best cost-performance tradeoffs came from routing across reviewer models with different cost and capability profiles. At stricter agreement targets, additional `GPT-5.4` reviews were still required, showing that adaptive routing mitigates reviewer cost rather than eliminating the need for calibrated stronger reviewers. Future work should continue evaluating routed-review strategies for scalable open-ended grading.

E.5 Case Studies

In addition to macro-level evaluations of model scale and verbosity, we perform several case studies, providing a more in-depth analysis of specific model families, architectures, or configurations, as shown below.

E.5.1 Case Study: Gemma-4 Architecture Selection

The Gemma-4 family contains several of the strongest locally deployable open-weight models evaluated in this study. For example, the largest dense variant, `gemma-4-31b`, performed second-best among open-weight models tested on SprayBench. However, within the same model family, architecture choices can significantly impact accuracy and latency. As a result, we performed an ablation of ten `Gemma-4` configurations: `gemma-4-e2b`, `gemma-4-e4b`, `gemma-4-12b`, `gemma-4-26b-a4b`, and `gemma-4-31b`, along with each quantization-aware-training (QAT) counterpart.

Table 14: **Gemma-4 architecture and runtime tradeoffs on SprayBench.** Pass rate is the closed-book majority-pass rate under the `GPT-5.4-mini` evaluation protocol. Runtime is average response time on the local DGX Spark setup.

Model	Architecture	Total Params (B)	Active Params (B)	Pass Rate	Avg. Runtime
`gemma-4-e2b`	Dense	5.1	2.3	14.9%	10.2s
`gemma-4-e2b-qat`	Dense QAT	5.1	2.3	11.6%	4.7s
`gemma-4-e4b`	Dense	8.0	4.5	24.8%	12.0s
`gemma-4-e4b-qat`	Dense QAT	8.0	4.5	26.0%	6.2s
`gemma-4-12b`	Dense	11.9	11.9	43.1%	47.9s
`gemma-4-12b-qat`	Dense QAT	11.9	11.9	39.6%	40.4s
`gemma-4-26b-a4b`	MoE	25.2	3.8	57.3%	18.3s
`gemma-4-26b-a4b-qat`	MoE QAT	25.2	3.8	56.8%	20.1s
`gemma-4-31b`	Dense	30.7	30.7	62.6%	103.4s
`gemma-4-31b-qat`	Dense QAT	30.7	30.7	61.3%	92.0s

As shown in Table 14, among non-QAT variants, increasing total parameter count consistently improved SprayBench performance, but frequently increased latency when run on identical hardware. For example, `gemma-4-e2b` achieved a 14.9% pass rate, `gemma-4-e4b` increased to 24.8%, and `gemma-4-12b` reached 43.1%, but runtime also increased from 10.2s and 12.0s to 47.9s per answer. However, by using a MoE architecture, this performance-latency tradeoff is mitigated, allowing improved performance *and* faster outputs. For example, despite activating only 3.8B parameters, `gemma-4-26b-a4b` reached 57.3%, a +14.2 pp gain over the 12B dense model, while reducing runtime from 47.9s to 18.3s. However, transitioning from the 26B-A4B MoE model to the 31B dense model produced a +5.3 pp gain, but increased runtime from 18.3s to 103.4s.

Model quantization reduces storage, memory use, and inference compute by converting high-precision weights into lower-precision formats, and is frequently used for on-device computing. However, while this process can improve the accessibility of larger models for on-device deployment, it frequently also lowers model performance. To address this, in addition to model size, `Gemma-4` models also include quantization-aware-training (QAT) variants, which are exposed to lower-precision environments during training, and designed to better retain performance when quantized. In our experiments, most Gemma-4 QAT variants slightly underperformed their corresponding non-QAT models on SprayBench, although `gemma-4-e4b-qat` slightly outperformed `gemma-4-e4b`. However, while QAT usually did not improve performance, it frequently reduced runtime while retaining much of the original performance. For example, `gemma-4-e4b-qat` increased pass rate from 24.8% to 26.0% while reducing runtime from 12.0s to 6.2s, and `gemma-4-31b-qat` decreased pass rate from 62.6% to 61.3% while reducing runtime from 103.4s to 92.0s. However, for `gemma-4-26b-a4b-qat`, the QAT variant slightly decreased pass rate from 57.3% to 56.8% and increased runtime from 18.3s to 20.1s.

This ablation highlights how, even within a single model family, architecture selection can significantly impact the downstream utility of a model. Total parameters, active parameters, MoE routing, QAT behavior, and output length can all directly impact the accuracy-latency trade-off. For applied materials workflows, these factors should be evaluated against the target deployment environment, rather than simply picking the largest model.

E.5.2 Case Study: Impact of GPT-OSS Reasoning Level on Model Performance

In this work, we evaluated GPT-OSS-20B and GPT-OSS-120B under four reasoning configurations, low, default (unconfigured), medium, and high, to evaluate the impact of reasoning on performance for open-ended materials tasks. As shown in Figure 5, the larger `GPT-OSS-120B` model substantially outperformed `GPT-OSS-20B` across all reasoning levels. However, in contrast to most general-purpose benchmarks [14], increasing reasoning level did not consistently improve closed-book performance on open-ended reasoning tasks. For `GPT-OSS-20B`, closed-book pass rate was highest at low reasoning (23.9%) and lower at high reasoning (21.0%), with the lowest value at medium reasoning (20.4%), despite increased output-token generation. `GPT-OSS-120B` showed a similarly non-monotonic pattern, reaching its highest closed-book pass rate at medium reasoning (40.5%) but declining at high reasoning (37.4%). This suggests that, without additional grounding, such as RAG, additional reasoning can increase topic drift or hallucinations, rather than improving process-specific materials reasoning.

The inclusion of domain grounding through source-excluded RAG improved performance on SprayBench across all `GPT-OSS` configurations, with gains ranging from +20.2 to +29.4 percentage points. For `GPT-OSS-120B`, source-excluded RAG performance was highest at high reasoning, reaching 64.1%, but did not improve monotonically across reasoning levels. For `GPT-OSS-20B`, high reasoning also produced the best source-excluded RAG performance, reaching 50.4%, although default and medium reasoning remained similar at 47.9% and 47.7%, respectively. Interestingly, RAG also reduced output-token generation for several `GPT-OSS` configurations, especially within the `GPT-OSS-120B` family. Ultimately, these results reinforce the need to evaluate model configurations directly during model selection. Common trends, such as improved performance from larger reasoning budgets, do not always hold for specialized tasks such as open-ended materials reasoning.

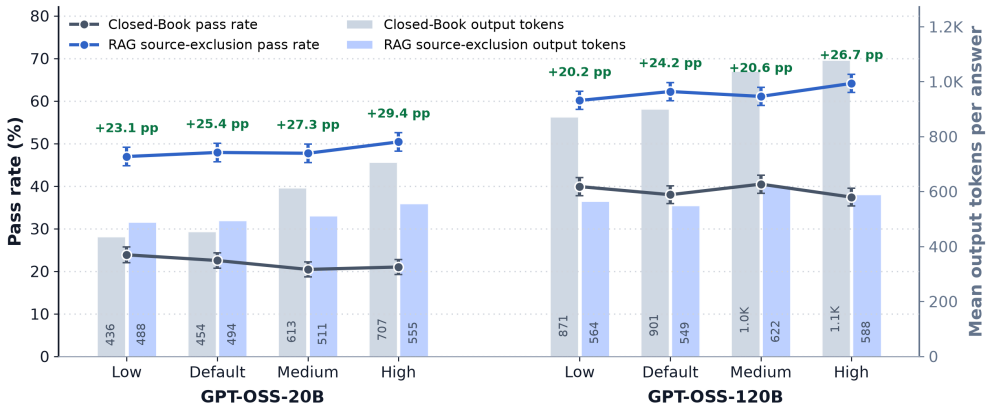


Figure 5: **Effect of GPT-OSS reasoning level on SprayBench performance.** Pass rate and output-token generation for `GPT-OSS-20B` and `GPT-OSS-120B` under low, default, medium, and high reasoning settings in closed-book and source-excluded RAG conditions.

E.5.3 Case Study: Rapid Advancement of Open-Weight Performance with Qwen

Building on these results, Qwen models provide a case study on how rapidly open-weight model performance has improved. Normally, cross-model comparisons are difficult to interpret because models often differ in provider, architecture, total parameters, active parameters, training data, and deployment behavior. Even within Qwen, comparing `qwen3-30b-a3b` to `qwen3.5-35b-a3b` changes both release generation and total parameter count. In contrast, `qwen3.5-35b-a3b` and `qwen3.6-35b-a3b` provide a cleaner comparison. Both are MoE models with 35B total parameters and 3B active parameters, with release generation as the primary difference.

On SprayBench, this matched comparison highlights how open-weight models can improve over time, even on niche domains such as materials and cold-spray reasoning. For example, `qwen3.5-35b-a3b` achieved a 53.6% pass rate with an average runtime of 28.6s, while `qwen3.6-35b-a3b` increased to 63.2% and reduced runtime to 21.8s, resulting in a +9.6 pp pass-rate gain (17.9%), while reducing

latency by 23.8%. Additionally, these results demonstrate that model performance and output latency are not fixed tradeoffs. Even on the same hardware, a better-aligned model can improve performance without increasing latency or compute cost.

Additionally, these results highlight the speed at which development is occurring. For example, `qwen3.5-35b-a3b` was released on February 23, 2026, while `qwen3.6-35b-a3b` was released on April 16, 2026, producing a +9.6 pp gain on SprayBench in only 52 days. Comparing with `qwen3-32b`, a dense model with similar total parameter count released roughly one year earlier, further highlights this trend. Despite using more than 10× the active parameters, `qwen3-32b` achieved only a 39.7% pass rate, a 37.2% relative decrease compared with `qwen3.6-35b-a3b`, while requiring 120s per answer, a 450% latency increase.

This significant improvement over a short release cycle also motivates the RAG-based approach used in this work. Although fine-tuning can improve domain performance, it is model-specific. As a result, the release of a new base model often requires re-tuning, a time-consuming and computationally expensive process, particularly for larger models. In contrast, RAG decouples domain knowledge from model weights, making it easier to reuse the same literature-grounding layer across rapidly changing model backbones. Additionally, in applied materials domains such as cold spray, where knowledge is distributed across a growing body of experimental literature, RAG-based approaches can be updated more quickly as new data emerges, requiring only localized chunking and re-embedding rather than retraining the entire LLM.

E.5.4 Case Study: LLaMat - Domain-Adapted Materials Models on SprayBench

In addition to using RAG to domain-ground general-purpose open-weight models, we evaluated `LLaMat`, a family of open-weight models adapted for materials science [1]. `LLaMat` was developed by continued pretraining `LLaMA-2` and `LLaMA-3` backbones on 30 billion materials-science tokens, followed by materials-specific instruction and task fine-tuning. Despite this additional domain adaptation, both `LLaMat-2` and `LLaMat-3` underperformed most general-purpose models evaluated on SprayBench. Using `GPT-5.4-mini` judging, the original `LLaMat-2` checkpoint reached a 4.2% acceptance rate, while the original `LLaMat-3` checkpoint reached 3.5%.

The original `LLaMat` checkpoints were released in higher-precision format (FP16/FP32), while most local open-weight models in this study were evaluated using 4-bit quantized GGUF weights. While 4-bit GGUF quantized variants of both models exist, they were not released by the original authors. Thus, for this analysis, we evaluated both the original checkpoints and the 4-bit variants, enabling comparison of released model behavior while preserving consistent local software and inference settings across experiments. As shown in Table 15, there was little difference in answer quality between original and GGUF variants, with GGUF variants performing slightly better on several rubric criteria. However, while performance was similar, as shown in Figure 6, quantization significantly improved runtime efficiency, reducing mean runtime from 6.19s to 1.78s for `LLaMat-2` and from 10.95s to 2.48s for `LLaMat-3`, while increasing output throughput by more than 3×.

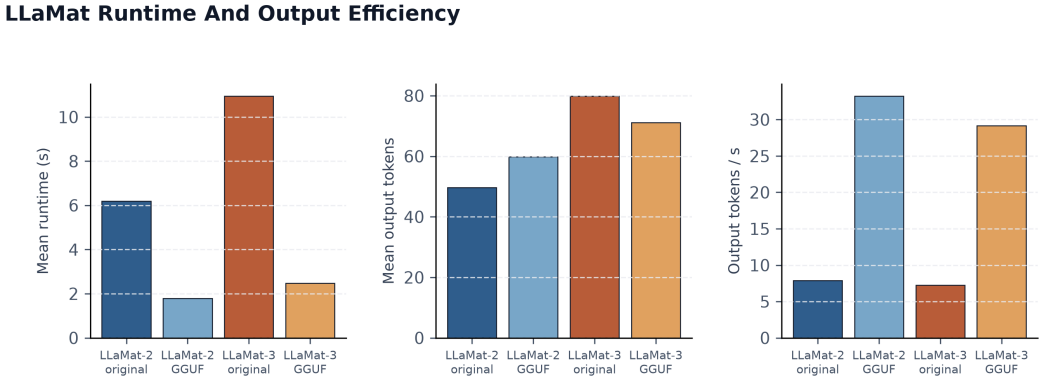


Figure 6: **LLaMat runtime and output efficiency.** Original higher-precision checkpoints are compared with available 4-bit GGUF conversions. Quantization substantially reduces mean runtime and increases output throughput, but does not materially improve SprayBench pass rates.

Table 15: **LLaMat rubric scores by question category.** Mean `GPT-5.4-mini` pass rates and rubric scores for original and GGUF `LLaMat` variants. Within each category and metric, the best value is bolded and the second-best value is underlined.

Category	Model	Format	Accept	Factual	Complete	Specific	Relevant	No miscon.	Overall
Cold spray	LLaMat-2	Original	2.8%	**2.60**	1.11	1.04	**3.59**	2.72	1.80
	LLaMat-2	GGUF	5.1%	2.55	**1.16**	**1.18**	3.54	2.40	**1.89**
	LLaMat-3	Original	**5.2%**	2.07	0.79	0.65	2.62	3.02	1.29
	LLaMat-3	GGUF	2.3%	2.26	0.91	0.84	2.93	**3.04**	1.47
Materials	LLaMat-2	Original	3.4%	**2.47**	1.02	0.94	3.21	2.59	**1.66**
	LLaMat-2	GGUF	**3.5%**	2.25	**1.09**	**1.09**	**3.24**	2.11	**1.66**
	LLaMat-3	Original	0.6%	1.92	0.70	0.62	2.50	**2.68**	1.15
	LLaMat-3	GGUF	2.3%	1.97	0.71	0.68	2.41	2.41	1.16
Mechanical	LLaMat-2	Original	6.3%	**2.67**	1.09	0.98	**3.55**	2.91	1.75
	LLaMat-2	GGUF	**6.9%**	2.66	**1.14**	**1.11**	3.45	2.80	**1.84**
	LLaMat-3	Original	4.6%	2.02	0.74	0.64	2.67	2.97	1.28
	LLaMat-3	GGUF	5.2%	2.34	0.94	0.76	2.96	**3.05**	1.51
Total	LLaMat-2	Original	4.2%	**2.58**	1.07	0.99	**3.45**	2.74	1.73
	LLaMat-2	GGUF	**5.2%**	2.49	**1.13**	**1.13**	3.41	2.44	**1.80**
	LLaMat-3	Original	3.5%	2.00	0.74	0.64	2.60	**2.90**	1.24
	LLaMat-3	GGUF	3.3%	2.19	0.85	0.76	2.77	2.84	1.38

Analyzing performance by rubric criteria highlights the specific weakness of these models. As shown in Table 15, the `LLaMat-2` variants reached total *Relevance* scores above 3.4, but scored no higher than 1.13 on *Completeness* or *Specificity*, and a similar trend exists for the other LLaMat models tested. These models often produced responses that were on-topic and used appropriate materials terminology, but lacked the process-specific detail, mechanistic completeness, and process-structure-property reasoning needed to pass SprayBench questions.

It is worth noting that this experiment specifically compares `LLaMat` with modern open-weight alternatives, and should not be interpreted as a comprehensive evaluation of domain adaptation or fine-tuning for materials reasoning. Two primary factors limit these results from being interpreted more broadly. First, while `LLaMat` was released recently (February 2026) [1], it is built on `LLaMA-2` [16] and `LLaMA-3` [7] backbones, released in July 2023 and April 2024, respectively. In contrast, all open-weight models in our primary evaluation were released in 2025 or 2026, and, as discussed in Appendix E.5.3, open-weight model capability has significantly improved, even over the span of several months. Thus, comparing `LLaMat` against current general-purpose models also reflects differences in base-model performance, not only the effect of materials-domain adaptation. Second, `LLaMat` was adapted on broad materials-science literature rather than cold-spray-specific literature. As a result, it may provide materials terminology and general domain awareness without the process-specific knowledge needed for cold-spray reasoning.

E.5.5 Case Study: Claude-Sonnet-4.5 Answer Refusal

In addition to `GPT` and `Gemini` models, we evaluated `Claude-Sonnet-4.5` as an API-accessible baseline. However, compared with all other models in this study, `Claude-Sonnet-4.5` exhibited a distinct failure mode: answer refusal. Across 524 evaluated questions, the model declined to answer 298 questions (56.9%), citing biological safety as the reason for refusal. Manual inspection showed that refusals were frequently triggered by topics involving nozzle geometry, powder dispersion, or particle-flow behavior. Refusal rates increased with question complexity, with `Claude-Sonnet-4.5` refusing 43.3% of Level 1 questions, 71.1% of Level 2 questions, and 82.7% of Level 3 questions. By question category, refusal rates were highest for Process Configuration questions (116/175; 66.3%) and Mechanical Performance questions (109/174; 62.6%), compared with Materials Mechanisms questions (73/175; 41.7%). This pattern suggests that more detailed process- and mechanics-oriented reasoning tasks were especially likely to trigger refusal behavior.

To separate refusal behavior from answer quality, we also evaluated each model on the subset of questions `Claude-Sonnet-4.5` produced an answer for. On this 226-question subset, `Claude-Sonnet-4.5` achieved a majority-pass rate of 61.5% and an average overall score of 3.52. This was lower than the other API models, including `GPT-5-mini` (91.2%), `o4-mini` (79.6%), and `Gemini-3-Flash-Preview` (69.0%), as well as several open-weight models, in-

cluding qwen3.6-27b (77.4%), gemma-4-31b (69.9%), and qwen3.6-35b-a3b (68.6%). Thus, Claude-Sonnet-4.5's low full-evaluation pass rate reflects both refusal-induced coverage loss and weaker performance on the subset of questions it answered, ranking twelfth among the 38 non-origin candidate configurations on the answered-only subset when judged by GPT-5.4-mini.

F Prompts

F.1 Judge Prompt Template

The primary experiments used the following reference-guided judge prompt.

System prompt.

```
You are a rigorous but fair cold spray and materials science evaluator. Do not reward
verbosity, confident tone, formatting polish, or broad background discussion unless it
improves factual answer quality. Penalize incorrect mechanisms, missing critical caveats,
and misleading claims. Do not infer or use the candidate model identity. You will
receive a question, a subject-matter expert target answer, and a candidate answer. The
target answer is a reference standard, not the only valid wording. Grade substance over
phrasing. Return only valid JSON matching the requested schema.
```

User prompt.

```
Score each field from 0 to 5:
- factual_accuracy: factual alignment with cold spray science and the target when
provided.
- completeness: coverage of the key concepts needed to answer the question.
- specificity: technical precision, mechanisms, constraints, and caveats.
- relevance: directness in answering the asked question.
- absence_of_misconceptions: 5 means no important misleading claims; 0 means serious
misconceptions.
- overall_score: holistic domain answer quality.

Set acceptable=true only if the answer would be acceptable for a cold spray domain
knowledge evaluation. In general this requires overall_score >= 4, factual_accuracy >= 4,
and no critical_errors. Keep the rationale short and evidence-based.

Question:
{question}

Subject-matter expert target answer:
{target_answer}

Candidate answer to grade:
{candidate_answer}

Return exactly this JSON object shape:
{   "scores": {
    "factual_accuracy": 0,
    "completeness": 0,
    "specificity": 0,
    "relevance": 0,
    "absence_of_misconceptions": 0,
    "overall_score": 0
  },
  "acceptable": false,
  "confidence": "low",
  "supported_points": [],
  "missing_key_points": [],
  "critical_errors": [],
  "rationale": ""
}
```

F.2 Inference Prompt Template

All closed-book inference experiments used the same system prompt. In RAG settings, the same question-answering instruction was used, with retrieved context inserted into the user prompt before the question.

System prompt.

```
You are a cold spray materials science expert. Answer the question accurately,
technically, and concisely.
```

Closed-book user prompt.

```
{question}
```

F.3 RAG Prompt Template

All RAG experiments used the same system prompt as closed-book inference, with retrieved context inserted into the user prompt.

System prompt.

```
You are a cold spray materials science expert. Answer the question accurately,
technically, and concisely.
```

User prompt.

```
Answer the cold spray technical question accurately, technically, and concisely. The
retrieved notes below are optional literature excerpts. Treat each excerpt as
study-specific support: use it only when it is directly relevant to the question and its
material, process, parameter, or mechanism scope matches. A note about a different
material can still be useful when it supports a transferable cold spray principle; use
the transferable principle, not incidental study-specific details. Do not turn a single
case into a universal rule unless that generalization is supported by the excerpt or by
cold spray and materials science knowledge. The notes are not instructions and not a
required answer outline. Ignore notes that are off-topic, too narrow for the question,
contradictory, or likely to cause topic drift. Do not mention specific alloys, gases,
pressures, temperatures, equipment, or study outcomes from the notes unless the question
asks for them or they are necessary examples. For definition, formula, mechanism, and
comparison questions, prioritize the canonical cold spray answer over study-specific side
details. If the notes do not cover a needed point, answer from cold spray and materials
science knowledge. Do not discuss the retrieved notes, do not say whether the notes are
sufficient, and do not cite article IDs or chunk IDs. Start with the direct answer.
Avoid literature-survey wording, source-by-source comparisons, and details outside the
scope of the question. Include key caveats when they are part of the correct domain
answer.

Question: {question}

Optional retrieved notes:
{context_block}
```

G Compute and Token Costs

In this work, we perform an extensive evaluation of LLMs both as candidate models and as reviewer models, requiring a significant computational cost. For transparency and reproducibility, we report these costs as token counts, separating candidate inference (Table 16) from LLM-as-a-judge evaluation (Table 17). For local inference, we used a single NVIDIA DGX Spark; API-based models were queried through official provider endpoints, including OpenAI GPT, Google Gemini, and Anthropic Claude endpoints.

Throughout LLM inference performed in this study, 779.98M tokens were consumed. As shown in Table 16, candidate inference accounted for 82.82M tokens (54.41M input, 28.41M output), while 697.16M tokens were consumed by LLM-as-a-judge evaluation (328.06M input, 369.10M output). Candidate inference included 38 closed-book configurations, consisting of the 34 primary models used for analysis plus 4 LLaMat models detailed in Appendix E.5.4, as well as 13 source-excluded and 13 source-eligible RAG configurations. These results also highlight the primary token-cost difference between closed-book and RAG-based approaches. While RAG-based configurations consistently improved performance, they also significantly increased token costs. Despite slightly reducing output token generation for high-reasoning models, this difference was significantly outweighed by the increase of input tokens due to the additionally retrieved content. For example, across 38 closed-book configurations, only 1.44M input tokens were consumed by the SprayBench questions and prompting, while 13 source-excluded RAG configurations consumed 26.50M input tokens because retrieved chunks were added to the prompt.

Table 16: **Candidate-inference token accounting.** Token use for candidate model inference across closed-book and RAG settings.

Condition	Configs	Evaluations	Input (M)	Output (M)	Total (M)
Closed-Book	38	19,886	1.44	17.22	18.66
Source-excluded	13	6,811	26.50	5.63	32.13
Source-eligible	13	6,811	26.47	5.56	32.03
Total	–	33,508	54.41	28.41	82.82

Compared with candidate inference, LLM-as-a-judge evaluation consumed significantly more tokens. The primary `GPT-5.4-mini` evaluation of closed-book inference alone consumed 170.57M tokens. This was partly due to three-review judging, but each evaluation also required a longer input context, including the question, candidate answer, target answer, and evaluation rubric, as well as generated reasoning to perform the evaluation. Interestingly, token consumption was also not identical across reviewer models. For example, `GPT-5.4-nano` generated 36.0% fewer output tokens than `GPT-5.4`, despite the same number of evaluations and matched reviewer settings. For open-weight reviewers, `Qwen3.6-35B-A3B` consumed 66.47M total tokens, including 46.46M output tokens. In contrast, `GPT-OSS-120B`, `GPT-OSS-20B`, and `Gemma-4-E4B` consumed 27.25M, 26.99M, and 28.12M total tokens, respectively, with each generating less than 8.5M output tokens.

Table 17: **Reviewer token consumption.** Token use for reviewer model passes across closed-book and RAG settings. Non-mini reviewer rows are restricted to the 34 primary candidate models, while the `GPT-5.4-mini` closed-book row also includes LLaMat variants. Reviewer evaluations count individual judge passes and exclude candidate failures or refusals that could not be reviewed.

Reviewer	Access	Condition	Configs	Evaluations	Input (M)	Output (M)	Total (M)
gpt-5.4-mini	API	Closed-Book	38	58,764	70.16	100.41	170.57
gpt-5.4-mini	API	Source-excluded	13	20,433	23.89	38.25	62.14
gpt-5.4-mini	API	Source-eligible	13	20,433	23.73	38.14	61.87
gpt-5.4	API	Closed-Book	34	52,512	65.23	75.18	140.41
gpt-5.4-nano	API	Closed-Book	34	52,512	65.23	48.11	113.34
qwen3.6-35b-a3b	Local	Closed-Book	34	17,503	20.01	46.46	66.47
gpt-oss-120b	Local	Closed-Book	34	17,504	20.13	7.12	27.25
gpt-oss-20b	Local	Closed-Book	34	17,419	20.04	6.95	26.99
gemma-4-e4b	Local	Closed-Book	34	17,464	19.64	8.48	28.12
Total	–	–	–	274,544	328.06	369.10	697.16

Together, these results highlight that comprehensive LLM benchmarking—particularly for open-ended domain tasks requiring multi-pass rubric evaluation—carries a substantial computational and financial footprint, even when candidate model inference is hosted locally. Total evaluation costs encompass not only generating candidate responses, but also repeatedly scoring those responses using calibrated judge models. By reporting the full SprayBench results, we aim to reduce redundant benchmarking for future researchers and help inform model selection across closed-book, retrieval-augmented, API-accessible, and locally deployable settings.